

Challenge: Remote Control Relay

Points: 100 • Estimated Time: 60 Minutes

Challenge Scope

A student team must teleoperate the robot through a slalom course, swapping drivers at the halfway mark.

Instructions & Engineering Hints

- **Objective:** Your team must successfully teleoperate your robot through a challenging slalom course, executing a mandatory driver swap at the exact halfway mark to cross the finish line in the fastest cumulative time.
- **Architectural & Driving Tips:**
 - • **Ergonomics & Control:** Design your remote control layout or choose your drive team wisely. Precision matters more than raw speed when weaving through tight cones. Consider using a proportional drive setup in your code if available to make gentle turns smoother.
 - • **Visibility & Orientation:** Mount your control hub or any visual markers so the drivers can easily maintain spatial awareness, especially when the robot changes direction or when handing off the controller.
 - • **The Handoff Strategy:** Practice the driver swap meticulously. Designate a specific 'Pit Stop' zone where the first driver safely halts the robot, places the controller down, and the second driver takes over instantly. Time lost here is difficult to make up on the track.